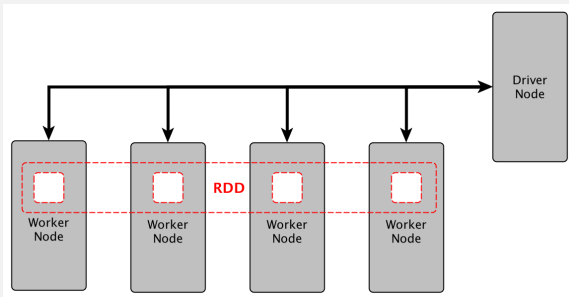


Spark RDDs



Resilient Distributed Datasets

- ▶ A **RDD** (Resilient Distributed Dataset) is a fault-tolerant collection of elements that can be operated on in parallel.
- ▶ There are two ways to create RDDs:
 - ▶ parallelizing an existing collection in your driver program
 - ▶ referencing a dataset in an external storage system, such as a shared filesystem, HDFS, HBase, or any data source offering a Hadoop InputFormat.

Create Spark Context

- ▶ First, we need to create a Spark context to start using spark. from a [Collection](#). For example:

- ▶ In Java:

```
import org.apache.spark.SparkConf;  
import org.apache.spark.api.java.JavaRDD;  
import org.apache.spark.api.java.JavaSparkContext;  
  
SparkConf conf = new SparkConf().setAppName("appName").  
    setMaster("local[*]");  
JavaSparkContext sc = new JavaSparkContext(conf);
```

- ▶ In Python:

```
from pyspark import SparkContext, SparkConf  
  
conf = SparkConf().setAppName("appName").setMaster("local[*]  
")  
sc = SparkContext(conf=conf)
```

RDDs from Existing Collection

- ▶ The `parallelize` method on the `JavaSparkContext` is used to create a RDD from a `Collection`. For example:

In Java:

```
List<Integer> data = Arrays.asList(1, 2, 3, 4, 5);
JavaRDD<Integer> distData = sc.parallelize(data); //uses
    default #partitions
JavaRDD<Integer> distData = sc.parallelize(data, 2); //use 2
    partitions
```

In Python:

```
data = [1, 2, 3, 4, 5]
distData = sc.parallelize(data) #uses default # of partitions
distData = sc.parallelize(data, 2) #uses 2 partitions
```

- ▶ Now we could use parallel operations on this RDD. For example:

In Java:

```
distData.reduce((a, b) -> a + b);
```

In Python:

```
distData.reduce(lambda a, b: a + b)
```

RDDs from External Datasets

- ▶ Spark can create distributed datasets from any storage source supported by Hadoop, including your local file system, HDFS, HBase, Amazon S3 and others.
- ▶ Spark supports text files, [SequenceFiles](#), and any other Hadoop [InputFormat](#).
- ▶ Text file RDDs can be created using SparkContext's `textFile` method. This method takes an URI for the file (either a local path on the machine, or a `hdfs://`, `s3a://`, etc URI) and reads it as a collection of lines. For example:
- ▶ In Java:

```
JavaRDD<String> distFile = sc.textFile("data.txt");  
//now, we can use parallel operators  
int totalLength = distFile.map(s -> s.length()).reduce((a, b) ->  
    a + b)
```

- ▶ In Python:

```
distFile = sc.textFile("data.txt")  
distFile.map(lambda s: len(s)).reduce(lambda a, b: a + b)
```

- ▶ All of Spark's file-based input methods, including `textFile`, support running on directories, compressed files, and wildcards as well.
- ▶ [JavaSparkContext.wholeTextFiles](#) method lets you read a directory containing multiple small text files, and returns each of them as (filename, content) pairs.
- ▶ [JavaRDD.saveAsObjectFile](#) and [JavaSparkContext.objectFile](#) support saving an RDD in a simple format consisting of serialized Java objects.
- ▶ The rest of the notes will use Java/Scala examples and then we will go through the Python examples live in class as they are very similar.

RDD Operations

- ▶ RDDs support two types of operations:
 - ▶ **transformation**: creates a new dataset from an existing one
 - ▶ **action**: returns a value to the driver program after running a computation on the dataset
- ▶ All transformations in Spark are **lazy**, in that they do not compute their results right away. Spark remembers the order of the transformations. This allows it be more efficient and fault tolerant.
- ▶ By default, each transformed RDD may be recomputed each time we run an action on it. However, we may also persist an RDD in memory using the **persist** (or **cache**) method. There is also support for persisting RDDs on disk, or replicated across multiple nodes.
- ▶ Example:

```
JavaRDD<String> lines = sc.textFile("data.txt");
JavaRDD<Integer> lineLengths = lines.map(s -> s.length());
// to persist
lineLengths.persist(StorageLevel.MEMORY_ONLY());
int totalLength = lineLengths.reduce((a, b) -> a + b);
```

Function Passing to Spark

- ▶ Spark's API relies heavily on passing functions in the driver program to run on the cluster. In Java, functions are represented by classes implementing the interfaces in the `org.apache.spark.api.java.function` package. This can be done in two ways:
 - ▶ We can create an anonymous or a named class that implements one of the defined function interfaces and pass an instance to Spark
 - ▶ Use lambda expressions to concisely define an implementation
 - ▶ Note that anonymous inner classes in Java can also access variables in the enclosing scope as long as they are marked final. Spark will ship copies of these variables to each worker node as it does for other languages.

The Way of Not the Lambda (1)

Define classes that implements special interface provided by Spark:

```
class GetLength implements Function<String, Integer> {  
    public Integer call(String s) { return s.length(); }  
}  
class Sum implements Function2<Integer, Integer, Integer> {  
    public Integer call(Integer a, Integer b) { return a + b; }  
}
```

```
JavaRDD<String> lines = sc.textFile("data.txt");  
JavaRDD<Integer> lineLengths = lines.map(new GetLength());  
int totalLength = lineLengths.reduce(new Sum());
```


The Way of Not the Lambda (2)

Use anonymous classes:

```
JavaRDD<String> lines = sc.textFile("data.txt");
JavaRDD<Integer> lineLengths = lines.map(new Function<String,
    Integer>() {
    public Integer call(String s) { return s.length(); }
});
int totalLength = lineLengths.reduce(new Function2<Integer,
    Integer, Integer>() {
    public Integer call(Integer a, Integer b) { return a + b; }
});
```

Function Passing to Spark (Python)

- ▶ Spark's API relies heavily on passing functions in the driver program to run on the cluster. There are three ways to do this in Python:
 - ▶ Lambda expressions, for simple functions that can be written as an expression.
 - ▶ Local `def`s inside the function calling into Spark, for longer code
 - ▶ Top-level functions in a module. Note that this requires sending the object that contains the class along with the method.

Closures

- ▶ The **closure** is those variables and methods that must be visible for the executor to perform its computations on the RDD. Prior to execution, Spark computes the task's closure. This closure is serialized and sent to each executor.
- ▶ Example 1: What is the value of the counter? For local mode? For local[2] mode? For a cluster?

```
int counter = 0;
JavaRDD<Integer> rdd = sc.parallelize(data); // data is
    [1,2,3,4]
```

```
// Wrong: Don't do this!!
rdd.foreach(x -> counter += x);
println("Counter value: " + counter);
```

- ▶ We would use an **Accumulator** for the above scenario.

```
LongAccumulator accum = sc().longAccumulator();

sc.parallelize(Arrays.asList(1, 2, 3, 4)).foreach(x -> accum
    .add(x));
// ...

accum.value();
// returns 10
```

Printing Elements of an RDD

```
//Scala version
rdd.foreach(println);
rdd.map(println);
// both of the above idioms won't work, why?

// use one of the below idioms (depending on the size of the
// RDD)
rdd.collect().foreach(println);
rdd.take(k).foreach(println); //print first k values

//Java version
rdd.collect().foreach(s -> System.out.println(s));
rdd.take(k).foreach(s -> System.out.println(s)); //print
first k values
```

Working with Key-Value Pairs

- ▶ Key-value pairs are represented by the `Tuple2` class (from Scala). For example: `tuple = new Tuple2(a, b)`. We can access the fields by `tuple._1()`, `tuple._2()`.
- ▶ RDDs of key-value pairs are represented by `JavaPairRDD` class. These are built using special versions of `map` operations such as `mapToPair` and `flatMapToPair`.
- ▶ Example: Count how many times each line of text occurs in a file.

```
JavaRDD<String> lines = sc.textFile("data.txt");
JavaPairRDD<String, Integer> pairs = lines.mapToPair(s
    -> new Tuple2(s, 1));
JavaPairRDD<String, Integer> counts = pairs.reduceByKey
    ((a, b) -> a + b);
// more examples: counts.sortByKey(), counts.collect();
```

Transformations (1)

- ▶ **map**(func): Return a new distributed dataset formed by passing each element of the source through a function func.

```
rdd = {1, 2, 3, 3}
rdd.map(x -> x + 1)  {2, 3, 4, 4}
```

- ▶ **filter**(func): Return a new dataset formed by selecting those elements of the source on which func returns true.

```
rdd = {1, 2, 3, 3}
rdd.filter(x -> x != 1)  {2, 3, 3}
```

- ▶ **distinct**([numPartitions]): Return a new dataset that contains the distinct elements of the source dataset.

```
rdd = {1, 2, 3, 3}
rdd.distinct()  {1, 2, 3}
```

- ▶ **flatMap**(func): Similar to map, but each input item can be mapped to 0 or more output items. For example:

```
x = {1, 2, 3, 3}
x = x.to(3) //generates a range from x up to 3
x = {{1, 2, 3}, {2, 3}, {3}, {3}}
rdd.flatMap(x) --> {1, 2, 3, 2, 3, 3, 3}
```

Transformations (2)

- ▶ **sample**(withReplacement, fraction, seed): Sample a fraction of the data, with or without replacement, using a given random number generator seed.
- ▶ **union**(otherDataset) Produce an RDD containing elements from both RDDs.

```
rdd = {1, 2, 3}   other = {3, 4, 5}
rdd.union(other)  {1, 2, 3, 3, 4, 5}
```

- ▶ **intersection**(otherDataset) Produce an RDD containing common elements from both RDDs.

```
rdd = {1, 2, 3}   other = {3, 4, 5}
rdd.intersection(other)  {3}
```

- ▶ **subtract**(otherDataset)

```
rdd = {1, 2, 3}   other = {3, 4, 5}
rdd.subtract(other)  {1, 2}
```

- ▶ **cartesian**(otherDataset)

```
rdd = {1, 2, 3}   other = {3, 4, 5}
rdd.cartesian(other)  {(1, 3), (1, 4), (1, 5), ...,
                       (3, 5)}
```

Transformations (3)

Pair RDDs are allowed to use all the transformations available to standard RDDs. In addition, we have extra transformations that only apply to pair RDDs.

- ▶ **groupByKey**([numPartitions]): When called on a dataset of (K, V) pairs, returns a dataset of (K, Iterable<V>) pairs.

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.groupByKey()    {(1, [2]), (3, [4, 6])}
```

- ▶ **reduceByKey**(func, [numPartitions]): When called on a dataset of (K, V) pairs, returns a dataset of (K, V) pairs where the values for each key are aggregated using the given reduce function func, which must be of type (V,V) => V.

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.reduceByKey((a, b) -> a + b)    {(1, 2), (3, 10)}  
//default is to use add function
```

- ▶ **aggregateByKey**(zeroValue)(seqOp, combOp, [numPartitions]): It is designed to aggregate values for each key using a different logic for within-partition reduction (*SeqOp*) versus cross-partition (*CombOp*) merging using a neutral "zero" value that is an identity for the combine functions. This is useful for aggregating data of different types, for example to compute the average of values by key, where we have to keep track of both the sum and the count, i.e. we have to return a pair of numbers while computing the average.

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.aggregateByKey(0)((a,b) -> a + b), ((a,b) -> a + b))  
    {(1, 2), (3, 10)}
```


Transformations (4)

- ▶ **mapValues(func)** Apply function to each value of a pair without changing the key.

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.mapValues(x -> x + 1)    {(1, 3), (3, 5), (3, 7)}
```

- ▶ **flatMapValues(func)** Apply function (that returns an iterator) to each value of a pair. For element returned, produce a key/value pair entry with the old key. Useful for tokenization.

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.flatMapValues(x -> range(x, 5) )  
{(1, 2), (1,3), (1,4), (1,5), (3, 4), (3,5)}
```

- ▶ **sortByKey([ascending], [numPartitions])**

```
rdd = {(3, 4), (1, 2), (3, 6)}  
rdd.sortByKey()    {(1, 2), (3, 4), (3, 6)}
```

- ▶ **keys()** Return an RDD of just the keys

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.keys()    {1, 3, 3}
```

- ▶ **values()** Return an RDD of just the values

```
rdd = {(1, 2), (3, 4), (3, 6)}  
rdd.values()    {2, 4, 6}
```

Transformations (5)

- ▶ **subtractByKey**

```
rdd = {(1, 2), (3, 4), (3, 6)} other = {(3, 9)}  
rdd.subtractByKey(other)  
{(1, 2)}
```

- ▶ **join**(otherDataset, [numPartitions])

Also available: **leftOuterJoin**, **rightOuterJoin**,
fullOuterJoin

```
rdd = {(1, 2), (3, 4), (3, 6)} other = {(3, 9)}  
rdd.join(other)  
{(3, (4, 9)), (3, (6, 9))}
```

- ▶ **cogroup**(otherDataset, [numPartitions]) Group data from both RDDs sharing the same key

```
rdd = {(1, 2), (3, 4), (3, 6)} other = {(3, 9)}  
rdd.cogroup(other)  
{(1, ([2], []), (3, ([4, 6], [9])))}
```

Transformations (6)

Finally, some system level operations.

- ▶ `pipe(command, [envVars])`

Pipe each partition of the RDD through a shell command. RDD elements are written to the process's stdin and output to its stdout are returned as an RDD of strings (one per line of output).

- ▶ `coalesce(numPartitions)`

Decrease the number of partitions in the RDD to numPartitions. Useful for running operations more efficiently after filtering down a large dataset.

- ▶ `repartition(numPartitions)`

Reshuffle the data in the RDD randomly to create either more or fewer partitions and balance it across them. This always shuffles all data over the network.

- ▶ `repartitionAndSortWithinPartitions(partitioner)`

Repartition the RDD according to the given partitioner and, within each resulting partition, sort records by their keys. This is more efficient than calling repartition and then sorting because it can push the sorting down into the shuffle machinery.

Actions (1)

- ▶ `reduce(func)`: The function should be commutative and associative so that it can be computed correctly in parallel.
- ▶ `collect()`: Return all the elements of the dataset as an array at the driver program. Only useful for a small subset of the data.
- ▶ `count()`: Return the number of elements in the dataset.
- ▶ `first()`: Return the first element of the dataset (similar to `take(1)`).
- ▶ `take(n)`: Return an array with the first `n` elements of the dataset.
- ▶ `takeSample(withReplacement, num, [seed])`: Return an array with a random sample of `num` elements of the dataset, with or without replacement, optionally pre-specifying a random number generator seed.
- ▶ `takeOrdered(n, [ordering])`: Return the first `n` elements of the RDD using either their natural order or a custom comparator.

Actions (2)

- ▶ **saveAsTextFile**(path): Write the elements of the dataset as a text file (or set of text files) in a given directory in the local filesystem, HDFS or any other Hadoop-supported file system. Spark will call `toString` on each element to convert it to a line of text in the file.
- ▶ **saveAsSequenceFile**(path): Write the elements of the dataset as a Hadoop SequenceFile. This is available on RDDs of key-value pairs that implement Hadoop's Writable interface.
- ▶ **saveAsObjectFile**(path): Write the elements of the dataset in a simple format using Java serialization, which can then be loaded using `SparkContext.objectFile()`.
- ▶ **countByKey**(): Only available on RDDs of type (K, V). Returns a hashmap of (K, Int) pairs with the count of each key.
- ▶ **foreach**(func): Run a function `func` on each element of the dataset. This is usually done for side effects such as updating an Accumulator or interacting with external storage systems.

Shuffle Operations (1)

- ▶ **Shuffle** operation re-distributes the data so that it's grouped differently across partitions. Although the set of elements in each partition of newly shuffled data will be deterministic, and so is the ordering of partitions themselves, the ordering of these elements is not.
- ▶ Involves disk I/O, data serialization, and network I/O. Complex and costly.
- ▶ Shuffle can be triggered by operations like **repartition**, **coalesce**, *byKey* operations (except for counting) such as **groupByKey** and **reduceByKey**, and join operations like **cogroup** and **join**.

Performance of Shuffle

- ▶ The shuffle is implemented using map and reduce (similar to Hadoop MapReduce)
- ▶ Internally, results from individual map tasks are kept in memory until they can't fit. Then, these are sorted based on the target partition and written to a single file. On the reduce side, tasks read the relevant sorted blocks.
- ▶ Certain shuffle operations can consume significant amounts of heap memory since they employ in-memory data structures to organize records before or after transferring them. When data does not fit in memory Spark will spill these tables to disk, incurring the additional overhead of disk I/O and increased garbage collection.
- ▶ Shuffle also generates a large number of intermediate files on disk. These files are preserved until the corresponding RDDs are no longer used and are garbage collected.

- ▶ Work through this notebook: [RDD examples](#).

RDD Persistence

- ▶ Persisting (caching) allows future actions to be much faster (often more than 10x). Caching is a key tool for interactive algorithms and fast interactive use.

```
rdc.persist(StorageLevel.MEMORY_ONLY());
```

MEMORY_ONLY	Store in memory. Recompute as needed
MEMORY_AND_DISK	
MEMORY_ONLY_SER	
MEMORY_AND_DISK_SER	
DISK_ONLY	
MEMORY_ONLY_2	Replicate each partition on two nodes
MEMORY_AND_DISK_2	
OFF_HEAP	

- ▶ Spark automatically removes old data partitions in a LRU (Least Recently Used) fashion. We can manually remove an RDD with the `unpersist()` method on an RDD object.

Broadcast Variables

- ▶ Limited support for shared variables across workers.
- ▶ **Broadcast variables** allow the programmer to keep a read-only variable cached on each machine rather than shipping a copy of it with tasks. They can be used, for example, to give every node a copy of a large input dataset in an efficient manner.
- ▶ Spark automatically broadcasts the common data needed by tasks within each stage. The data broadcasted this way is cached in serialized form and deserialized before running each task. Thus explicitly creating broadcast variables is only useful when tasks across multiple stages need the same data or when caching the data in deserialized form is important.

- ▶ Example:

```
Broadcast<int[]> broadcastVar = sc.broadcast(new int[] {1,  
    2, 3});
```

```
broadcastVar.value();  
// returns [1, 2, 3]
```

- ▶ After the broadcast variable is created, it should be used instead of the original value in any functions run on the cluster. (Why?)
- ▶ The object should not be modified after it is broadcast in order to ensure that all nodes get the same value of the broadcast variable (e.g. if the variable is shipped to a new node later).

Built-in Accumulators

- ▶ **Accumulators** are variables that are only “added” to through an associative and commutative operation and can therefore be efficiently supported in parallel. They can be used to implement counters (as in MapReduce) or sums.
- ▶ Spark natively supports accumulators of numeric types, and programmers can add support for new types.
- ▶ We can create *named* or *unnamed* accumulators. Named accumulators will show in the web UI for Spark.
- ▶ Example:

```
LongAccumulator accum = sc.longAccumulator();

sc.parallelize(Arrays.asList(1, 2, 3, 4)).foreach(x -> accum
    .add(x));
// ...

accum.value();
// returns 10 (only readable on driver)
```

Derived Accumulators

- Accumulators for custom types can be created by subclassing `AccumulatorV2` abstract class.

```
class VectorAccumulatorV2 implements AccumulatorV2<MyVector,
    MyVector> {
    private MyVector myVector = MyVector.createZeroVector();

    public void reset() {
        myVector.reset();
    }

    public void add(MyVector v) {
        myVector.add(v);
    }
    ...
}

// Then, create an Accumulator of this type:
VectorAccumulatorV2 myVectorAcc = new VectorAccumulatorV2();
// Then, register it into spark context:
sc.register(myVectorAcc, "MyVectorAcc1");
```

Accumulator behaviors

- ▶ *Note:* A buggy accumulator will not impact a Spark job, but it may not get updated correctly although a Spark job is successful.
- ▶ For accumulator updates performed inside actions only, Spark guarantees that each task's update to the accumulator will only be applied once, i.e. restarted tasks will not update the value.
- ▶ In transformations, users should be aware of that each task's update may be applied more than once if tasks or job stages are re-executed.
- ▶ Accumulators do not change the lazy evaluation model of Spark. If they are being updated within an operation on an RDD, their value is only updated once that RDD is computed as part of an action. Consequently, accumulator updates are not guaranteed to be executed when made within a lazy transformation like `map()`.

```
LongAccumulator accum = sc.longAccumulator();  
data.map(x -> { accum.add(x); return f(x); });  
// Here, accum is still 0 because no actions have caused the  
  `map` to be computed.
```